

Astrodynamics I: Final Exam

June 13, 2011

1. A spacecraft is orbiting a planet in a circular orbit with a radius of 4 Planet Radii ($a=4$, $e=0$). A maneuver is performed that rotates the velocity vector in the orbital plane by an angle of 45° towards the planet.
 - A. Find the new semimajor axis a_n and the new eccentricity e_n following this maneuver.
 - B. At what point in its new orbit is the spacecraft in terms of true anomaly (v)?
 - C. Will the spacecraft impact the planet? Justify your answer.

2. A ballistic rocket is launched from the surface of the Earth into an orbit. After burnout the orbit is described in the orbital plane by

$$r = \frac{5000}{1 + 0.33 \cos(\theta + 136)} \text{ km,}$$

where r is the radial distance in km from the Earth's center to the rocket and θ is the anomaly (angle) measured in degrees from burnout. The radius of the Earth is $R_E=6,378\text{km}$ and the gravitational parameter of the Earth is $\mu_E=398,600\text{km}^3/\text{sec}^2$. Ignore the effect of the Earth's atmosphere.

- A. What is the altitude of the rocket at burnout? What is the velocity of the rocket at burn out?
 - B. What is the maximum altitude of the rocket? What is the anomaly at which this maximum altitude occurs? What is the velocity when the rocket reaches its maximum altitude?
 - C. What is the anomaly at which the rocket impacts the Earth? What is the distance, along a great circle on the Earth's surface that the rocket travels from burn out to impact? What is the velocity of the rocket when it impacts the Earth?
3. (Conversion of Orbital Elements to state vectors)
 - A. Show that the position and velocity vector in the perifocal coordinate system is:

$$\vec{r}_{PQW} = \begin{bmatrix} \frac{pcosv}{1+ecosv} \\ \frac{psinv}{1+ecosv} \\ 0 \end{bmatrix} \text{ and } \vec{v}_{PQW} = \begin{bmatrix} -\sqrt{\frac{\mu}{p}} \sin v \\ \sqrt{\frac{\mu}{p}} (e + \cos v) \\ 0 \end{bmatrix}$$

- B. For the following orbital elements:

$$p = 0.23DU, e = 0.82, i = 90^\circ, \Omega = 180^\circ, \omega = 260^\circ, \nu = 190^\circ$$

determine $\vec{r}_{PQW}$ and $\vec{v}_{PQW}$. Note: set $\mu = 1$.

- C. By a suitable coordinate transformation, express $\vec{r}$ and $\vec{v}$ in the geocentric equatorial system in the $\hat{I}, \hat{J}, \hat{K}$ axis.
4. Given $\vec{r}_{IJK}, \vec{v}_{IJK}, \vec{r}_{siteIJK}, \phi_{gd}, \theta_{LST}$, describe the procedure to compute range (ρ), azimuth (β), elevation (el), and their time rates.
5. Discuss the following terminology:
 - A. Geocentric Latitude & Geodetic Latitude
 - B. Ecliptic & Obliquity
 - C. Apparent Solar Time, Mean Solar Time, Universal Time & Equation of Time
 - D. Canonical Units